Renjie Lyu

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I am an assistant professor at the School of Mathematical Sciences, Xiamen University. Previously, I was a postdoctoral researcher at the Morningside Center of Mathematics, Chinese Academy of Sciences. I received my Ph.D. from the University of Amsterdam under the supervision of Mingmin Shen.
My research is in algebraic geometry, with interests in Hodge theory, cubic hypersurfaces, algebraic cycles, and automorphisms of algebraic varieties.

Employment

- Sep. 2023–present. Assistant Professor, School of Mathematical Sciences, Xiamen University.
- Oct. 2020–Aug. 2023. Postdoctoral Researcher, Morningside Center of Mathematics, Chinese Academy of Sciences. Mentor: Baohua Fu.

Education

- Sep. 2016–Oct. 2020. Ph.D. in Mathematics, University of Amsterdam.
  Supervisor: Mingmin Shen.
- Aug. 2014–May 2016. M.A. in Mathematics, Washington University in St. Louis.
- Sep. 2010–July 2014. B.S. in Mathematics, Zhejiang University.

Publications and Preprints

Published Articles

- Renjie Lyu, *Cylinder maps of algebraic cycles on cubic hypersurfaces*, *Bulletin of the London Mathematical Society* **57** (2025), no. 12, 3949–3968. DOI; arXiv:1810.12394.
- Renjie Lyu and Xuanyu Pan, *Remarks on automorphism and cohomology of finite cyclic coverings of projective spaces*, *Mathematical Research Letters* **28** (2021), no. 3, 785–822. DOI; arXiv:1705.06618.

Accepted Article

- Renjie Lyu and Zhiwei Zheng, *Degenerations to secant cubic hypersurfaces and limiting Hodge structure*, to appear in *Algebraic Geometry*. arXiv:2206.12670.

Preprints

- Renjie Lyu and Xiping Zhang, *Remarks on the monodromy conjecture for symmetric determinantal hypersurfaces*, 2026. arXiv:2610.18395.
- Renjie Lyu and Xuanyu Pan, *Limiting Hodge Structures for Cubic Hypersurfaces of Secant Type*, 2026. arXiv:2607.11198.
- Wenfei Liu and Renjie Lyu, *The Chevalley–Weil formula for finite group actions on higher dimensional compact complex manifolds*, 2025. arXiv:2510.10430.
- Renjie Lyu and Dingxin Zhang, *Generic Triviality of Automorphism Groups of Complete Intersections*, 2024. arXiv:2404.11037.
- Renjie Lyu, *Lines on the secant cubic hypersurfaces of Severi varieties*, 2021. arXiv:2112.00335.

Doctoral Thesis

- Renjie Lyu, *Algebraic cycles on cubic hypersurfaces and Fano scheme of lines*, University of Amsterdam, 2020. Thesis.

Invited Talks

- *Degeneration of Hodge structures and cubic hypersurfaces,* Purdue Algebraic Geometry Seminar (online), Purdue University, Sep. 2023.
- *Degeneration of Hodge structures and cubic hypersurfaces,* Moduli Spaces and Related Topics, YMSC, Tsinghua University, June 2023.
- *Generic Triviality of automorphisms on algebraic varieties,* Xiamen University, June 2023.
- *Degeneration of Hodge structures and cubic hypersurfaces,* International Young Scholars Forum, Tongji University, Jan. 2023.
- *Degeneration of Hodge structures and cubic hypersurfaces,* The 2nd Global Youth Forum, Institute of Geometry and Physics, USTC, Dec. 2022.
- *Degeneration of cubic hypersurfaces and limiting Hodge structure,* Shanghai Jiao Tong University, Aug. 2022, online.
- *Degeneration of cubic hypersurfaces and limiting Hodge structure,* 3rd National Algebraic Geometry Conference, Fudan University, July 2022, online.
- *Lines on the secant cubic hypersurfaces of Severi varieties,* MCM Members Seminar, CAS, Nov. 2021.
- *Cubic hypersurfaces: Chow groups, motives and rationality problems,* MCM Members Seminar, CAS, Dec. 2020.
- *The automorphism and cohomology of cyclic coverings,* Séminaire Géométrie de l'ICJ, Université Claude Bernard Lyon 1, May 2019.
- *A result of algebraic cycles on cubic hypersurfaces,* Rationality of Algebraic Varieties, Schiermonnikoog Island, Apr. 2019.
- *A result of algebraic cycles on cubic hypersurfaces,* Algebraic and Arithmetic Seminar, University of Amsterdam, Feb. 2019.
- *A result of algebraic cycles on cubic hypersurfaces,* Forschungsseminar "Arithmetische Geometrie", Humboldt University, Jan. 2019.
- *A result of algebraic cycles on cubic hypersurfaces,* Algebraic Geometry Seminar, Chinese Academy of Science, Dec. 2018.
- *Chow rings of hyper-Kähler manifolds,* GAeL XXV, University of Bath, June 2017.

Honors and Awards

- S.-T. Yau College Student Mathematics Contest 2013,
  Geometry and Topology, Bronze Medal.

Seminar Talks

- *Characterization of cubic hypersurfaces in projective geometry*, preprint seminar, MCM, Mar 2022.
- *Degeneration of Hodge structures*, preprint seminar, MCM, Nov 2021.
- *Lagrangian fibration on irreducible symplectic manifolds*, Lagrangian Fibration Seminar, MCM, Mar 2021.
- *Fano variety of lines on cubic fourfolds*, Reading Seminar on Cubic Hypersurfaces, MCM, Jan 2021.
- *Fqpc sites: descent theory*, Étale Cohomology Seminar, University of Amsterdam, Sep 2019.
- *Bounding the trace of monodromy representations*, Reading Seminar on Finiteness of Monodromy Representation, University of Amsterdam, Nov 2018.
- *Deligne-Mumford stacks*, Moduli Stacks Seminar, Utrecht, Oct 2017.
- *Isogenous K3 surfaces are twisted derived equivalent*, Algebraic and Arithmetic Seminar, University of Amsterdam, Sep 2017.
- *Tate modules and fundamental groups of abelian varieties*, Abelian Varieties Seminar, Feb 2017.
- *Basics of abelian varieties*, Abelian Varieties Seminar, Oct 2016.

Seminars (Co-)Organized

- Lagrangian Fibration and Hyper-Kähler variety, MCM, Spring 2021.
- Seminar on Good Reduction of Abelian Varieties, University of Amsterdam, Fall 2016.

Teaching Experience

Teaching Assistantships

- Differential Geometry, Fall 2019.
- Abelian Variety, Fall 2018.
- Algebra 2, Fall 2018.
- Topology, Spring 2018.
- Algebraic geometry 1, Fall 2017.
- Topology, Spring 2017.
- Representation Theory, Fall 2016.
- Introduction to Lebesgue Integration, Spring 2016.
- Complex Analysis II, Spring 2016.
- Complex Variables, Fall 2015.
- Geometry I, Fall 2015.
- Complex Analysis I, Fall 2015.